

Table 1. Machine Learning Model Accuracy Test Results (Non-Quantize).

Trials	TP	TN	FP	FN	Number of Fruit	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
1	6	4	0	0	10	100	100	100	70,00
5	6	4	0	0	10	100	100	100	70,00
10	6	4	0	0	10	100	100	100	80,00
15	6	4	0	0	10	100	100	100	90,00
20	6	3	1	0	10	100	85.71	92.31	60,00
25	4	6	0	0	10	100	100	100	70,00
30	4	6	0	0	10	100	100	100	80,00
1-30	145	145	4	6	300	96.03	97.32	96.67	96.67

Table 2. Machine Learning Model Accuracy Test Results (Quantize).

Trials	TP	TN	FP	FN	Number of Fruit	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
1	2	6	0	2	10	50.00	100	66.67	80,
5	4	4	0	2	10	66.67	100	80.00	80
10	1	4	0	5	10	16.67	100,	28.57	50
15	3	4	0	3	10	50.00	100,	66.67	70
20	1	4	0	5	10	16.67	100	28.57	50
25	2	4	0	4	10	33.33	100	50.00	60
30	2	5	0	3	10	40.00	100	57.14	70
1-30	79	146	3	72	300	52.32	96.34	67.81	75.00

Table 3. Data Transmission Speed Test Results to End Device (Non-Quantize and Quantize).

Non-Quantize		Quantize	
Trials	Data Transmission Time (ms)	Trials	Data Transmission Time (ms)
5	666.8	5	283.39
10	616.79	10	250.05

15	633.46	15	266.72
20	666.8	20	300.06
25	633.46	25	283.39
30	666.8	30	300.06
Average	644.02	Average	273.29

Table 4. Comparison of Non-Quantize and Quantize Model Performance.

Model	Accuracy (%)	Data Transmission Time (ms)
Non-Quantize	96.67	644.02
Quantize	75	273.29

Table 5. VPN Stability Test Results.

Day	Time	Minimum Ping (ms)	Maximum Ping (ms)	Average Ping (ms)	Packet Loss (%)
Saturday	08:12 – 19:15	39	135	52	0
Thursday	08:15 – 19:17	17	153	43	0
Friday	08:12 – 19:15	36	153	58	0